

|          |                            |
|----------|----------------------------|
| EXP.NO-2 | EDA-Data Import and Export |
|          |                            |

**AIM:**

To import data from **CSV, Excel, and web sources** using Pandas, handle different data formats, and export a DataFrame to an Excel file.

**PROCEDURE:**

1. Import the **Pandas** library.
2. Read the CSV file using `pd.read_csv()`.
3. Read the Excel file using `pd.read_excel()`.
4. Display the first few records using `head()`.
5. Import tabular data from a webpage using `pd.read_html()`.
6. Select the required table from the webpage.
7. Store the data in a Pandas DataFrame.
8. Export the DataFrame to an Excel file using `to_excel()`.
9. Verify the exported data using `head()`.

**PROGRAM:**

```
import pandas as pd

df_csv = pd.read_csv("/content/test_Y3wMUE5_7gLdaTN.csv")

print("CSV Data:")

print(df_csv.head())

df_excel = pd.read_excel(

    "/content/Historicalinvesttemp.xlsx",

    sheet_name="Sheet1")
```

```
print("\nExcel Data:")

print(df_excel.head())

url = "https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)"

tables = pd.read_html(url)

df_web = tables[0]

print("\nWeb Data:")

print(df_web.head())

df = pd.read_excel("/content/Historicalinvesttemp.xlsx")

df.to_excel("/content/weeeeeee.xlsx", index=False)

print("\nData exported successfully!")

print(df.head())
```

OUTPUT:

| PassengerId | Survived | Pclass | Name | Sex                                               | Age    | SibSp | Parch | Ticket | Fare             | Cabin   | Embarked |   |
|-------------|----------|--------|------|---------------------------------------------------|--------|-------|-------|--------|------------------|---------|----------|---|
| 0           | 1        | 0      | 3    | Braund, Mr. Owen Harris                           | male   | 22.0  | 1     | 0      | A/5 21171        | 7.2500  | NaN      | S |
| 1           | 2        | 1      | 1    | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0  | 1     | 0      | PC 17599         | 71.2833 | C85      | C |
| 2           | 3        | 1      | 3    | Heikkinen, Miss. Laina                            | female | 26.0  | 0     | 0      | STON/O2. 3101282 | 7.9250  | NaN      | S |
| 3           | 4        | 1      | 1    | Futrelle, Mrs. Jacques Heath (Lily May Peel)      | female | 35.0  | 1     | 0      | 113803           | 53.1000 | C123     | S |
| 4           | 5        | 0      | 3    | Allen, Mr. William Henry                          | male   | 35.0  | 0     | 0      | 373450           | 8.0500  | NaN      | S |

| Unnamed: 0 | Unnamed: 1 | Unnamed: 2              | Unnamed: 3 | Unnamed: 4 | Unnamed: 5 | Unnamed: 6      |
|------------|------------|-------------------------|------------|------------|------------|-----------------|
| 0          | NaN        | Excel Sample Data       | NaN        | NaN        | NaN        | NaN             |
| 1          | NaN        | NaN                     | NaN        | NaN        | NaN        | NaN             |
| 2          | NaN        | Students Marksheet Data | NaN        | NaN        | NaN        | NaN             |
| 3          | NaN        | NaN                     | NaN        | NaN        | NaN        | NaN             |
| 4          | NaN        | NaN                     | NaN        | NaN        | NaN        | Total Marks 300 |

|   | Student Name   | Exam Name | Date       | Points |
|---|----------------|-----------|------------|--------|
| 0 | James Walker   | Biology   | 05/06/2016 | 31     |
| 1 | Velma Clemons  | Biology   | 05/06/2016 | 43     |
| 2 | Kibo Underwood | Biology   | 05/06/2016 | 81     |
| 3 | Louis Mcgee    | Biology   | 05/06/2016 | 11     |
| 4 | Phyllis Paul   | Biology   | 05/06/2016 | 18     |

| PassengerId | Survived | Pclass | Name | Sex                                               | Age    | SibSp | Parch | Ticket | Fare             | Cabin   | Embarked |   |
|-------------|----------|--------|------|---------------------------------------------------|--------|-------|-------|--------|------------------|---------|----------|---|
| 0           | 1        | 0      | 3    | Braund, Mr. Owen Harris                           | male   | 22.0  | 1     | 0      | A/5 21171        | 7.2500  | NaN      | S |
| 1           | 2        | 1      | 1    | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0  | 1     | 0      | PC 17599         | 71.2833 | C85      | C |
| 2           | 3        | 1      | 3    | Heikkinen, Miss. Laina                            | female | 26.0  | 0     | 0      | STON/O2. 3101282 | 7.9250  | NaN      | S |
| 3           | 4        | 1      | 1    | Futrelle, Mrs. Jacques Heath (Lily May Peel)      | female | 35.0  | 1     | 0      | 113803           | 53.1000 | C123     | S |
| 4           | 5        | 0      | 3    | Allen, Mr. William Henry                          | male   | 35.0  | 0     | 0      | 373450           | 8.0500  | NaN      | S |

**RESULT:**

Thus, the given program was successfully executed to import data from CSV, Excel, and web sources and export a DataFrame to an Excel file.